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ORIGINAL ARTICLE

Zero-shot realistic image deblurring with consistency model

Zhaohan Wang[1] · Chengjun Chen[1] · Chenggang Dai[1]

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Abstract

At present, diffusion-based image deblurring methods rely on paired blurry-clear datasets for training, and the types of blur causes in image synthesis cannot yet be determined with sufficient precision to model real-world scene blur datasets. Furthermore, diffusion models are computationally expensive and may generate distort images during deblurring. To solve the above problems, this paper proposes image deblurring with consistency distillation models (ID-CDM) to restore blurred images in real scenes without training for image deblurring tasks. Specifically, this paper proposes a diffusion coefficient-dominated score model (DCSM), which ignores the impact of the drift term on the forward stochastic differential equation (SDE) adding Gaussian noise, thus simplifying the deblurring network architecture and reducing the computational complexity. The score model is integrated as a distillation model into the consistency model, aiming to enhance the quality of image deblurring while significantly reducing image sampling time. Besides, this paper employs the method of invertible linear transformation to highly compress the blurred image into the latent space for zero-shot processing, and iterative sampling is utilized to improve image accuracy and reduce additional artifact details generated by deblurring images. Finally, to verify the effectiveness of the proposed method, comparative analyses were conducted with classical deblurring methods on the synthetic blurring dataset HIDE and the real scene blurring dataset RealBlur. Analysis results indicate that ID-CDM is a highly competitive deblurring scheme.

Keywords Image deblurring · Image restoration · Diffusion model · Zero-shot image processing

Introduction

The image deblurring technology in real scenes has attracted much attention in image processing, which aims to recover a clear image from a blurred input image. Image blurring is caused by camera shake, out-of-focus images, fast-moving objects, and a mixture of multiple causes [1–3]. The traditional image deblurring method usually takes simplified ideal conditions as a preconditioning assumption and estimates the blur kernel that causes image blur. However, when the degraded model is insufficient to describe real data or the model estimation is poor, the image deblurring effect is unsatisfactory. In recent years, deep learning has been widely applied to image processing, among which convolutional neural networks [4–9] and transformers [10–13] have achieved great success. Meanwhile, the deblurring model evolved from transformers can capture long-distance dependencies of image data, which improves the processing ability of non-uniform blurred features. However, for complex blurring scenes of real images, this type of method cannot effectively process high-frequency details in blurred images [14].

Recently, diffusion models [15–18] have demonstrated excellent performance in the field of image generation and have been widely used in image restoration and computer vision. The generation process of the diffusion model is an iterative sampling process that progressively eliminates noise from random initial vectors. Compared with other models such as GAN (generative adversarial networks) [19], VAEs (variational auto-encoder) [20, 21], and normalizing flow [22, 23], diffusion models can generate more accurate target distributions without problems of unstable optimization distributions and model collapse. However, compared to single-step generation models, the iterative generation process of diffusion models usually requires about 10–2000 times more computation than the sample generation, which incurs huge computational and time costs, leading to slow

✉ Chengjun Chen
chencj@qut.edu.cn

[1] Qingdao University of Technology, Qingdao 266520, Shandong, China

inference speed and limitations in real-time applications. The consistency model proposed by Song [24] and other researchers [25–29] aims to reduce the sampling time of diffusion models. In addition, the image deblurring task requires adding precise details to the input blurred image. If the diffusion model is simply applied to the deblurring task, it will consume considerable computational resources and generates additional artifact that do not exist in the clear image, resulting in image distortion.

To solve the above problems, this paper proposes ID-CDM (image deblurring with consistency distillation models) for image deblurring under for both synthetic and real-world blurred datasets under zero-shot conditions. Zero-shot refers to training ID-CDM to guide image generation by learning the scores of clean target images under different noise levels during the distillation stage, without involving blurred images at any point. The visualization results of ID-CDM are shown in Fig. 1. Considering the numerous iterative steps and long sampling time of the diffusion model, this paper will conduct image deblurring experiments based on the consistency distillation model to reduce the image deblurring sampling time. Meanwhile, this paper designs a score network dominated by diffusion coefficients based on U-Net time correlation, which will be integrated into ID-CDM as a distillation model, and performs image generation experiments. Additionally, considering the high computational cost incurred by the image space, this paper will use the method of invertible linear transformation to map the blurred image into the latent space for compression, thereby effectively reducing the computational complexity. To sum up, compared with existing CNN-based, GAN-based, transformer-based, and diffusion-based deblurring approaches, the novelties of our proposed ID-CDM can be summarized as follows:

1. This study designs a diffusion coefficient-dominated score model (DCSM) that eliminates the drift term in stochastic differential equations and is dominated by the diffusion coefficient. This simplification reduces computational complexity, avoids numerical instability, and maintains accurate score estimation, which addresses the limitations of existing score-based diffusion models in terms of efficiency and stability.
2. This paper innovatively applies the consistency model to image deblurring. While consistency models enable few-step generation, their direct application to image restoration often leads to insufficient detail recovery. By integrating DCSM into the consistency framework, ID-CDM achieves a better balance between fast inference and fine-detail preservation, overcoming the trade-off faced by conventional consistency-based approaches.
3. This paper employs the method of invertible linear transformation to map blurred images into a compact latent space for zero-shot image deblurring experiments. This reduces the computational complexity of the model and avoids additional artifacts that do not exist in the given clear image, achieving generalization in complex blurred scenes.

Taken together, these contributions provide a new perspective for combining diffusion-based score modeling with consistency learning, offering a practical and efficient solution for real-world image deblurring.

Related works

With the development of deep learning, various deblurring schemes based on convolutional neural networks have been proposed. Nah et al. [7] developed a dynamic deblurring method based on deep multi-scale convolutional neural networks, which utilizes a multi-scale network structure to iteratively extract multi-scale information to restore images; Tao et al. [31] presented a scale recursive network model with shared parameters, which samples the weights of the shared network for training, increasing the stability of the model. GANs [19] have been widely used in image generation, image restoration, and other fields. DeblurGAN [32] introduces GAN into image deblurring and uses paired datasets for adversarial training. The results show that the restoration effect is better than that of convolutional neural network-based deblurring methods, but the method is still difficult to restore specific texture structures. Based on this, the research team proposed DeblurGAN-v2 [33], which uses a spatial pyramid network as the core generator and selects networks of different magnitudes as the backbone network of the pyramid, obtaining superior deblurring effects than DeblurGAN. DBGAN [30] combines two GAN models, blurring GAN (BGAN) learning and deblurring GAN (DBGAN) learning. BGAN learns how to blur clear images with unpaired sets of clear and blurred images, and then it guides DBGAN to learn how to correctly deblur these images. This method substantially improves the quality of image deblurring, but training two generative adversarial networks leads to unstable training and is prone to pattern collapse. Moreover, such methods often encounter limitations in practical applications [34], leading to suboptimal performance.

The diffusion model, as a new branch of generative models, has made a series of breakthroughs in generative tasks. Essentially, it is a probabilistic generative model that can construct required data samples from Gaussian noise through a random iterative denoising process. The prototype of the diffusion model can be traced back to article [15], and it was jointly developed and investigated by DDPM [16], IDDPM [35], and SDE (stochastic differential equation) [18]. Compared with GAN [19], diffusion models have advantages such as high fidelity and diverse types of generated

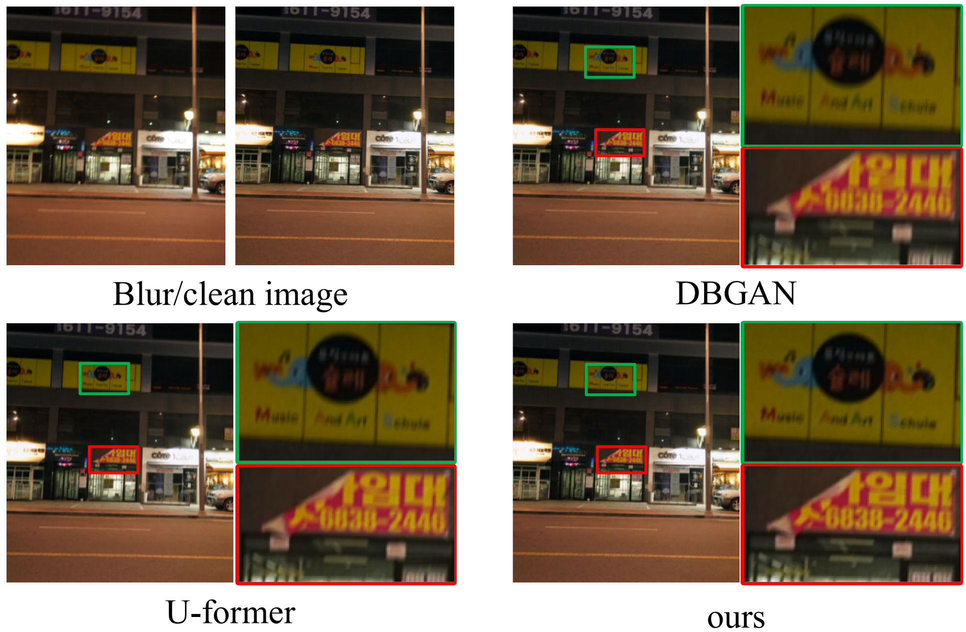

Fig. 1 The proposed method by making a comparison with the deblurring methods DBGAN [30] and U-former [11]

images, so they have successfully replaced GAN models in various fields such as visual generation [17, 29, 36] and conditional visual generation [37–40]. According to training strategies, image deblurring methods can be roughly divided into two categories: the first category [41–44] is dedicated to optimizing the diffusion model of image deblurring through supervised learning, and the other category is dedicated to using pre-trained deblurring diffusion models to generate prior information under zero-shot [45–49]. Methods based on supervised learning require collecting many blur/clear datasets, while zero-shot-based methods mainly rely on known blur forms. In real scenes, the causes of blur are usually diverse and unknown. Some studies [50–54] optimized diffusion models and combined them with real scenes to restore images. However, one of the defects of diffusion models is that numerous iterative steps are required during the sampling process. Chen et al. [55] replaced the noise schedule, Song et al. [29] changed the sampling strategy, and Rombach et al. [56] executed diffusion models in latent space, all of which aimed to reduce the sampling steps of diffusion models. Despite various attempts, the complexity of generating diffusion models is still high, especially for high-resolution images in image restoration. Additionally, image restoration methods based on diffusion models are prone to producing additional artifacts that do not exist in the clear image, so these methods do not perform well in terms of image distortion.

Considering the limitations of diffusion models in the field of image restoration, this paper designs an independent score model as the distillation model using consistency models for image deblurring, aiming to enhance the quality of image deblurring while significantly reducing image sampling time. Meanwhile, due to the high computational cost of the pixel space, this paper performs image deblurring in low dimensional latent space.

Background

Score-based diffusion model

In a diffusion model, data are converted to noise by Gaussian perturbation, and samples are created from noise by sequential denoising, thereby achieving data generation. In this study, the diffusion process is described by the SDE in the score model [18]. The forward diffusion can be expressed by:

$$dx = f(x, t)dt + g(t)dw \tag{1}$$

where x denotes i.i.d. data samples, dx represents the diffusion increment of each step, $f(\cdot, \cdot)$ refers to the drift coefficient, $g(\cdot)$ is the diffusion coefficient, w denotes standard Brownian motion, $t \in (0, T)$ denotes forward diffusion moment, and T refers to prior distribution moment. The reverse diffusion of SDE can be expressed by:

$$dx = \left[f(x, t) - g^2(t)\nabla logp_t(x)\right]dt + g(t)d\overline{w} \tag{2}$$

As the distribution of x at moment t can be expressed as $p_t(x)$, $p_0(x)$ and $p_T(x)$ are expressed as raw and prior data distributions, respectively; $\nabla logp_t(x)$ denotes the score of $p_t(x)$. Figure 2 shows a schematic diagram of the forward and Reverse SDE.

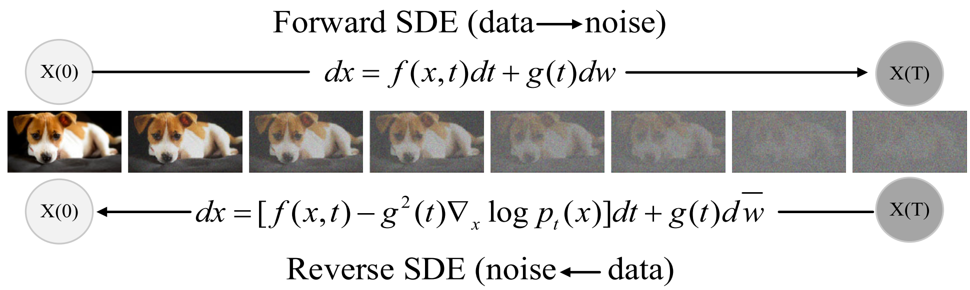

Fig. 2 Schematic diagram of the forward and reverse SDE

A prominent property of SDE is the existence of a probability flow ordinary differential equation (PF ODE), and samples obtained from the reverse SDE and PF ODE from the prior distribution belong to the same data distribution. The solution trajectories of the reverse diffusion of PF ODE at moment t can be determined by:

$$dx_t = \left[f(x,\, t) - \frac{1}{2}g^2(t)\nabla_x logp_t(x)\right]dt \tag{3}$$

Generally, the prior distribution in Eq. (1) is designed in such a way that $p_T(x)$ approximates a tractable Gaussian distribution. Let $f(x,\, t) = 0,\ g(t) = \sqrt{2t}$. Then, a score model $s_\phi(\cdot,\, \cdot)$ is trained to approximate the score of $p_t(x)$ by score matching methods [17], i.e., $s_\phi(x,\, t) \approx \nabla_x logp_t(x)$. Substituting it into Eq. (1), the empirical PF ODE is obtained:

$$\frac{dx_t}{dt} = -ts_\phi(x_t,\, t) \tag{4}$$

As the prior noise distribution satisfied $\widehat{x}_T \sim N(0,\, T^2 I)$, a numerical ODE calculator is employed to obtain the solution trajectories of Eq. (4). Finally, the estimated raw data distribution $\widehat{x}_0$ is obtained, which is considered as an approximate sample of the raw data distribution sample x_0.

Consistency model

To date, great advances have been achieved in the reduction of sampling time by diffusion models [24–29], e.g., The consistency model enables high-quality image generation in very few sampling steps. The consistency function is given below:

$$f : (x_t,\, t) \mapsto x_\varepsilon \tag{5}$$

where ε is a very small positive number.

Equation (3) provides the solution trajectory of sampling by PF ODE at moment t. T steps of iterative denoising are required to generate an image. The output of the consistency function for any pair $(x_t,\, t)$ belonging to the same PF ODE trajectory is consistent, i.e.,

$$\forall t,\, t\prime \in [\varepsilon,\, T],\ f(x_t,\, t) = f(x_{t\prime},\, t\prime) \tag{6}$$

The consistency neural network f_θ is constructed to learn Eq. (6). It means that any point at any time step in the consistency function can be mapped to the starting point of the trajectory.

Consider a deep neural network $F_\theta(x,\, t)$ without specific structural constraints (the output of this network has the same dimensionality as x), namely

$$f_\theta(x,\, t) = c_{skip}(t)x + c_{out}(t)F_\theta(x,\, t) \tag{7}$$

where $c_{skip}(\cdot)andc_{out}(\cdot)$ are differentiable functions. When $t = \varepsilon$, we have $c_{skip}(\varepsilon) = 1,\ c_{out}(\varepsilon) = 0$, i.e., $f_\theta(x,\, \varepsilon) = 1 \cdot x_\varepsilon + 0 \cdot F_\theta(x_\varepsilon,\, \varepsilon) = x_\varepsilon$, indicating that the boundary conditions of the consistency model are satisfied. The parameterization in Eq. (7) is very similar to various diffusion models [26, 57], making it easier to leverage powerful diffusion model architectures to establish the consistency model.

The trained consistency model f_θ can generate samples by sampling from the prior distribution $\widehat{x}_T \sim N(0,\, T^2 I)$. Meanwhile, the consistency model can generate samples in a single step, which greatly overcomes the defects of many iterative steps and a long iteration time for diffusion models. Additionally, the consistency model can generate images by iteratively adding and removing noises alternately, which enhances the quality of generated samples.

Method

As a type of generative model, diffusion models have achieved remarkable results in image synthesis and restoration (e.g., super-resolution [14, 58–60], restoration [56, 61, 62], and denoising [46]). Nevertheless, the iterative denoising process of diffusion models requires a large amount of computation and time, which is one of the main limitations of diffusion models in practical training. Recently, many studies have focused on accelerating the sampling process while improving the quality of obtained samples [17, 24–27, 63–75]. Herein, the consistency model [24] enables single-step generation during iterative sampling and allows for iterative generation that balances sample quality with computational efficiency. Also, the consistency model can improve the quality of generated images by alternatively removing and injecting noises. In this study, the consistency distillation model is utilized to investigate image deblurring by using only clear image datasets for model training. Based on this, the ID-CDM is proposed, which integrates DCSM as a distillation model into the consistency model. The iterative sampling of ID-CDM can improve the quality of generated images, and more importantly, significantly overcome the defects of diffusion models in terms of many steps, long iteration time, and high computational costs for sample generation. Overall, this study applies the ID-CDM method to image deblurring under zero-shot conditions for both synthetic blurred datasets and real-world blurred datasets, which involves compressing blurred images into a latent space using reversible linear transformations. This can avoid the problem of producing mismatched details with given clear images by diffusion-based image deblurring methods while reducing model complexity and enhancing the generalization of image deblurring in complex scenes.

This section introduces DCSM as a distillation model and proposes the ID-CDM model and the ID-CDM model for restoring blurred images under zero-shot conditions.

ID-CDM

As shown in Fig. 3, the ID-CDM method consists of two stages.

In the first stage, a pre-trained diffusion model is distilled. Herein, a DCSM-based model is proposed and employed as the distillation model for pre-training. Different from EDM [26] (a distillation network model is set), DCSM does not involve any drift term $f(\cdot, \cdot)$ and it is a network architecture entirely dominated by an exponential diffusion term $g(\cdot)$. Therefore, the computational complexity of DCSM is significantly reduced. Figure 3a shows the forward SDE of DCSM. Herein, the distillation network adopts a U-Net-like [76] and time-dependent score network structure, with time and prior distribution $x(T)$ as inputs. The time information is not simply embedded into the network but concatenated with the image in the format $[\sin(2\pi wt); \cos(2\pi wt)]$, $w \sim N(0, s^2 I)$ through a Gaussian random feature [77], a special form of time encoding. The ID-CDM distillation network is utilized to estimate the score of each piece of data in the prior distribution, i.e., $\nabla logp_t(x)$. In this case, the output of the distillation network is multiplied by $1/\sqrt{E[\|\nabla_x logp_t(x(t)|x(0))\|_2^2]}$. This not only guarantees that the network output is close to the score of $p_t(x)$ but also makes the l_2 norm of the network output close to the l_2 norm of $\nabla_x logp_t(x(t)|x(0))$, thereby ensuring the accuracy of the score network $s_\phi(x, t)$ in estimating the data score $\nabla logp_t(x)$. The proposed DCSM-based distillation model is more compatible with the consistency model, contributing to better knowledge distillation performance.

In the second stage, the pure Gaussian noise image $x_{t_{n+1}}$ is first obtained from the clear image I_{GT} through forward diffusion process, then the obtained noise image $x_{t_{n+1}}$, the moment t_{n+1} corresponding to the Gaussian noise, and the score $\nabla logp_t(x)$ obtained from the first-stage distillation model of the ID-CDM method are input into the numerical calculator $ODEsolver$. The numerical calculator estimates the data distribution $\widehat{x}_{t_n}^{\phi}$ at the previous moment t_n. In this way, two sets of adjacent data $\left(x_{t_{n+1}}, t_{n+1}\right)$ and $(\widehat{x}_{t_n}^{\phi}, t_n)$ are obtained, which are input into the consistency model for error minimization, thereby obtaining a single generated image I_{gc}. By setting the number of iterations N, the image quality can be improved through iterative sampling. Finally, the generated image I_{GC} of raw data is obtained.

DCSM

This paper proposes a score model called DCSM that does not have a drift term and is dominated by an exponential diffusion coefficient. In existing diffusion models, the dynamics are governed by both a drift term and a diffusion term. However, the drift term primarily acts as a low-frequency bias that shifts the data distribution without contributing to high-frequency detail recovery, which is critical in deblurring tasks. Moreover, since deep neural networks can inherently absorb such low-frequency shifts during training, the drift term becomes redundant. This simplification reduces model complexity and avoids instability arising from balancing drift and diffusion dynamics. Empirically, the absence of the drift term does not hinder convergence; instead, it accelerates training and improves stability.

This paper assumes an isotropic Gaussian forward process with negligible drift, i.e., $f(x, t) \approx 0$. In this case, the reverse dynamics reduce to estimating only the scaled score term:

$$dx \approx -g(t)^2 \nabla_x logp_t(x)dt + g(t)d\overline{w} \tag{8}$$

Therefore, the denoising network only needs to predict the score $\nabla_x logp_t(x)$, modulated by the diffusion coefficient $g(t)$. This greatly simplifies the parameterization compared with full drift–diffusion formulations, while retaining theoretical consistency. More General Case, the DCSM model is taken as the pre-trained distillation model for ID-CDM. In forward SDE diffusion as shown in Eq. (1), let $f(x, t) \approx 0$, $g(t) = \xi^t$, and then DCSM is obtained:

$$dx \approx \xi^t dw, \ t \in [\varepsilon, 1] \tag{9}$$

In this case, distillation pre-training is performed based on the score model in Eq. (9). Thus, the distribution after

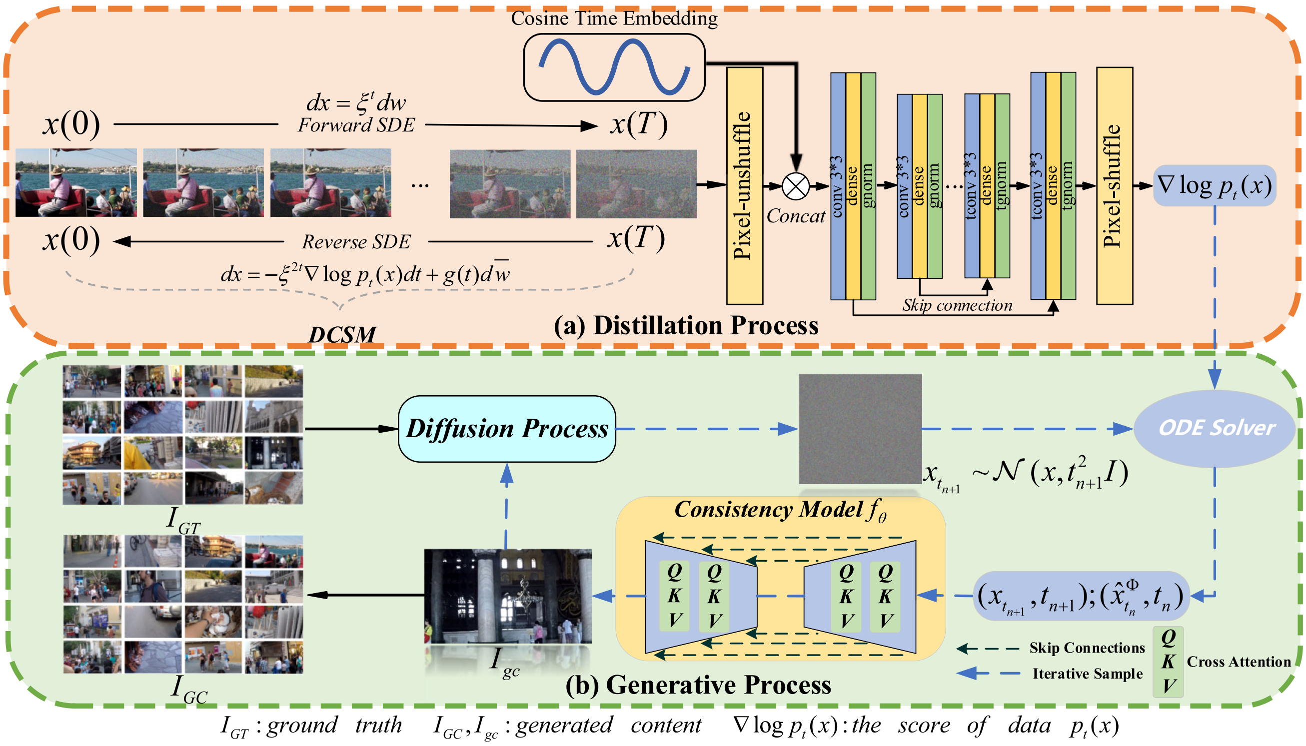

Fig. 3 The architecture of the ID-CDM method

DCSM forward diffusion process is:

$$p_t(x(t)|x(0)) = N\left(x(t);\, x(0),\, \left(\xi^{2t} - 1\right)I/2log\xi\right) \tag{10}$$

Since $p_t(x(t)|x(0))$ follows a Gaussian distribution, the weight $\lambda(t)$ is the variance of the data distribution. We also have:

$$\nabla_x logp_t(x(t)|x(0)) = -(x(t) - x(0))/\lambda(t) \tag{11}$$

Through parameter renormalization, there is:

$$x(t) = x(0) + \sqrt{\lambda(t)} \cdot z,\, z \sim N(0,\, I) \tag{12}$$

In Eq. (3), the objective function of training the score network $s_\phi(x,\, t)$ in the diffusion model is to estimate the score of $p_t(x)$ at different moments, i.e., $s_\phi(x,\, t) \approx \nabla_x logp_t(x)$. Its score loss function is:

$$\min_\theta E_{t\sim U(0,T)}[\lambda(t)E_{x(0)\sim p_0(x)}E_{x(t)\sim p_t(x(t)|x(0))}[\|s_\phi(x,\, t) \\ -\nabla_x logp_t(x(t)|x(0))\|_2^2]] \tag{13}$$

where t follows a uniform distribution, and $\lambda(t)$ is a weight coefficient used to balance the magnitudes of the score-matching loss functions at different moments.

If $p_t(x)$ follows a Gaussian distribution, then $\lambda(t)$ equals its variance. By substituting Eq. (12) into Eq. (11) and according to the loss function in Eq. (13), we have:

$$\begin{aligned} &\lambda(t)\|s_\phi(x(t),\, t) - \nabla_x logp_t(x(t)|x(0))\|_2^2 \\ &= \|s_\phi(x(t),\, t) \cdot \sqrt{\lambda(t)} + z\|_2^2, \\ &z \sim N(0,\, I) \end{aligned} \tag{14}$$

Then, the objective function for DCSM is:

$$\min_\theta E_{t\sim U(0,T)}[E_{x(0)\sim p_0(x)}E_{x(t)\sim p_t(x(t)|x(0))}\|s_\phi(x(t),\, t)\cdot \\ \sqrt{\left(\xi^{2t} - 1\right)/2log\xi} + z\|_2^2] \tag{15}$$

where $s_\phi(x,\, t)$ serves as the estimation model for the data score $\nabla_x logp_t(x)$ in the distillation stage of ID-CDM.

The DCSM is based on a U-Net-like architecture, consisting of multiple down-sampling and up-sampling layers, both of which contain convolutional operations with ReLU activation functions. The model uses skip connections to maintain feature maps across different layers and enable the flow of gradient information during training. Additional, to integrate time-dependent information, a Gaussian random feature encoding mechanism is employed. Time steps are encoded and concatenated with image features at each layer to enable the diffusion process over time. In each layer, the

DCSM estimates the score of the current image feature by calculating the gradient of the denoised image at that specific time step, allowing the model to progressively restore image details. Each layer of the DCSM uses 2D convolutions with kernel size 3×3, followed by batch normalization and ReLU activation. The final layer outputs a score map that is used to guide the image generation process.

ID-CDM for image generation

The DCSM is pre-trained to learn the score mapping without the drift term. This process helps the model efficiently estimate the denoising gradients. In the image generation stage, the consistency model uses the learned score map to progressively denoise and restore the image in a few steps, ensuring fast inference while maintaining image quality.

The DCSM is introduced into Eq. (4) as the distillation model. Herein, the distribution at moment t can be described by [26], i.e.:

$$t_i = \left(T^{1/\rho} + \frac{i-1}{N-1}\left(\varepsilon^{1/\rho} - T^{1/\rho}\right)\right)^{\rho} \tag{16}$$

The diffusion moment $t_i \in [\varepsilon, T]$ is discretized into N-1 sub-intervals, i.e., $t_1 = \varepsilon$, $t_N = T$, and ρ is a set constant. In the image generation stage, this study employs two methods, namely, the Euler sampler and the PF ODE numerical sampler, to directly compute the trajectory of Eq. (4) in a single step. Herein, the sampling process of the Euler sampler can be implemented through the pseudo-code in Algorithm 1. Literature [18] shows that the Euler numerical sampler and Langevin dynamic sampling [78, 79] can be combined for sampling. However, the combined method will lead to a slow sample generation speed, and the image quality generated by the combined method is similar to that using the Euler sampler only. Thus, the combined method is not used in this study. Another method is the PF ODE numerical sampler, which can accurately calculate the log-likelihood of the score function and has trajectories with the same marginal probability density as SDE. Additionally, the sampling samples of SDE and PF ODE have the same distribution. The iteration formula for the PF ODE is:

$$x_{t-\Delta t} = x_t + \Delta t \tau(x_t, t; \phi) \tag{17}$$

where $\tau(\cdot, \cdot; \phi)$ is the update function of the ODE solver, which is related to the score model. When t from 0 to T is divided into N intervals, both Algorithm 1 and Eq. (14) can accurately estimate $x_{t-\Delta t}$ from x_t.

Algorithm 1 Euler sampling

Input: The score model $s_\phi(\cdot,\cdot)$, the sequence of time points $t_1 > t_2 > \cdots > t_N$, $t \sim U(0,T)$, the initial distribution $p_T \sim N(x; 0, \frac{1}{2log\xi}(\xi^2 - 1)I)$
$x_T = \sqrt{\frac{1}{2log\xi}(\xi^2 - 1)} \cdot z, z \sim N(0, I)$
$\Delta t \leftarrow d_t$
for $t = T$ **to** 1 **do**
  $x_{t-\Delta t} = x_t + \xi^{2t} s_\phi(x_t, t)\Delta t + \xi^t \sqrt{\Delta t} z_t, z_t \sim N(0, I)$
end for
Output: $\boldsymbol{x}$

In the image generation stage, the sample point x_0 is first collected from the clear dataset I_{GT}, and noises are added to x_0 through forward SDE, i.e., $x_{t_{i+1}} \sim N(x_0, t_{i+1}^2 I)$. Then, based on the Euler sampler or the discretization steps in Eq. (14), the data distribution $\widehat{x}_{t_i}^{\Phi}$ at moment x_{t_i} is estimated. Subsequently, the two adjacent data points $(\widehat{x}_{t_i}^{\Phi}, x_{t_{i+1}})$ on the trajectory of the empirical PF ODE in Eq. (4) are calculated. These two datasets are input into the consistency model to minimize their discrepancy through training. The objective function is:

$$\min_{\theta, \theta^-} E[\lambda(t_i) d(f_\theta(x_{t_{i+1}}, t_{i+1}), f_{\theta^-}(\widehat{x}_{t_i}^{\Phi}, t_i))] \tag{18}$$

where t_i is given in Eq. (16), $\lambda(t)$ is consistent with the loss function of the distillation model and serves as a weight coefficient to balance the magnitude of the score matching loss function at different moments, and $d(\cdot, \cdot)$ is the metric matrix.

More details on the training of the consistency model can be found in [24]. After obtaining the output of the consistency model, a single generated image is obtained. Meanwhile, the iterative sampling step N can be set to enhance the quality of image generation through iterative adding noise and noise reduction.

The DCSM, which provides the necessary information for the consistency model to guide the restoration process, plays a crucial role in estimating the scores at each time step. Unlike simple concatenation of existing techniques, ID-CDM is designed to leverage the strengths of both DCSM and consistency learning in a complementary manner. DCSM provides accurate and efficient score estimation without the instability of drift-based models. The consistency model uses these score maps to progressively generate images with fewer iterations, thus improving the overall efficiency. The output from the DCSM passes into the consistency model directly by this way, the ID-CDM can achieve both high-quality deblurring and reduced computational cost.

Zero-shot image deblurring

ID-CDM can restore blurred images without using blurred subjects and corresponding clear image datasets for model training. This is because, similar to the diffusion model, the consistency model also allows for editing data under zero-shot conditions, which provides a new feasible method for image deblurring. As illustrated in Fig. 4, this paper adopts the reversible linear transformation method, which can compress the blurred image I_B into a highly compact latent space, thereby significantly reducing computational complexity and improving sampling efficiency. Then, through forward diffusion process, a Gaussian pure noise distribution x is obtained. The obtained noise distribution x and the corresponding noise addition time t_i are input into the trained consistency model f_θ. f_θ serves as a denoising network to denoise the Gaussian pure noise, and the resulting image is mapped back to the pixel space through reversible linear transformation to obtain the deblurred image I_{db_n}. In addition, multiple iterations can be conducted to improve the image deblurring performance and thus obtain the deblurred image I_{DB}. Without using blurred subjects and corresponding clear image datasets for model training, ID-CDM not only reduces computational complexity but also reduces additional artifacts of deblurred images that do not exist in the given clear images in diffusion-based methods, leading to a better generalization of deblurring in complex blurred scenes.

The pseudo-code for the image deblurring task is shown in Algorithm 2. Herein, y represents the blur image; Λ denotes a binary bitmask, $\Lambda \in \{0,1\}$, and $\Lambda[i, j, k, l] = \{^{1, k\geq 1}_{0, k=0}$; $\odot$ denotes element-wise multiplication, and moment t is set according to Eq. (16).

Algorithm 2 Zero-shot image deblurring

1: **Input:** CDM $\boldsymbol{f_\theta}(\cdot,\cdot)$, a sequence of time points $t_1 > t_2 > \cdots > t_N$, blur image y, invertible linear transformation A, and binary image mask Λ
2: $y \leftarrow A^{-1}[(Ay) \odot (1 - \Lambda) + 0 \odot \Lambda]$
3: Sample $x \sim N(y, t_1^2 I)$
4: $x \leftarrow f_\theta(x, t_1)$
5: $x \leftarrow A^{-1}[(Ay) \odot (1 - \Lambda) + (Ax) \odot \Lambda]$
6: **for** $n = 2$ **to** N **do**
7: Sample $x \sim N(x, (t_n^2 - \varepsilon^2)I)$
8: $x \leftarrow f_\theta(x, t_n)$
9: $x \leftarrow A^{-1}[(Ay) \odot (1 - \Lambda) + (Ax) \odot \Lambda]$
10: **end for**
11: **Output:** $\boldsymbol{x}$

In ID-CDM based image deblurring, the invertible linear transformation A maps the input image from the pixel space $R^{h\times w\times 3}$ to the latent space $R^{h\prime\times w\prime\times d}$, enabling the subsequent diffusion model to learn in a lower-dimensional and more stable space. The key design is to ensure the invertibility of the transformation, so that the original resolution image can be reconstructed from the latent representation. This study define A as an invertible linear transformation composed of two steps: (1) spatial down-sampling to reduce the resolution from $h \times w$ to $h\prime \times w\prime$. $(h\prime, w\prime) = (h/s, w/s)$, in which s is the spatial down-sampling factor (e.g., s=4 for GoPro experiments with input resolution 256 × 256, resulting in a latent space size of 64 × 64). (2) A fixed orthogonal projection $P \in R^{3\times d}$ applied to each pixel vector. Specifically, for each location (i, j), the RGB vector $x[i, j, :] \in R^3$ is projected onto a d dimensional orthogonal basis. This method defines A as a reversible linear transformation, which is used to map images to a latent space, i.e., $A : x \in \mathbb{R}^{\boldsymbol{h}\times\boldsymbol{w}\times 3} \mapsto \boldsymbol{y} \in \mathbb{R}^{\boldsymbol{h}'\times\boldsymbol{w}'\times\boldsymbol{d}}$,

$$y[i, j, k] = \sum_{l=0}^{2} x[i, j, l]P[l, k] \tag{19}$$

where P is generated via QR decomposition and fixed throughout training. This ensures invertibility with $P^{-1} = P^T$. This choice is motivated by three factors: (1) orthogonal transforms preserve vector norms and thus avoid distortion in the latent space; (2) QR decomposition is computationally efficient and numerically stable; and (3) orthogonality ensures exact invertibility, which is critical for reconstructing the original image without information loss. The entire invertible linear transformation process is illustrated in Algorithm 3.

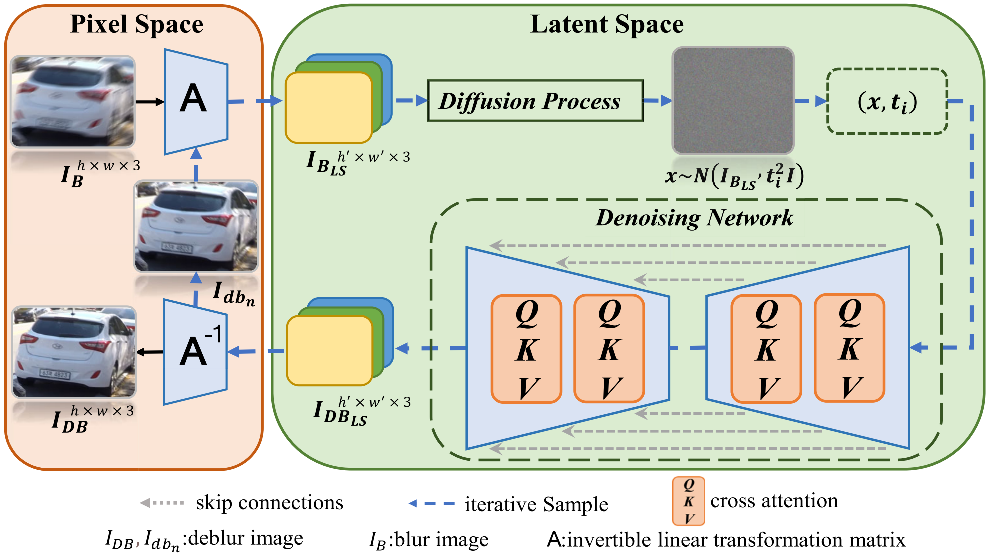

Fig. 4 The schematic of image deblurring by ID-CDM under zero-shot condition

Algorithm 3 Invertible latent space mapping

1: **Input:** Image $x \in R^{h\times w\times 3}$
2: $s = stride$, where s is the spatial down-sampling factor
3: $(h', w') = (h/s, w/s)$
4: **QR decomposition:** $P = QR(R), R \sim \mathrm{N}\ (0,1)^{3\times d}$
5: **Forward Transform:** $y[i,j,k] = \sum_{l=0}^{2} x[i,j,l]P[l,k], k = 0,1,\ldots,d-1$
6: $A: R^{h\times w\times 3} \to R^{h'\times w'\times d}, y = A(x) = Downsample(x)\cdot P$
7: **Inverse Transform:** $x[i,j,l] = \sum_{k=0}^{d-1} y[i,l,k]\, P^T[k,l]$
8: $(h, w) = (h' * s, w' * s)$
9: **Output:** $x \in R^{h\times w\times 3}$

In this study, the blurred image is processed under zero-shot conditions. First, the down-sampled image is preprocessed, and the blurred image is linearly transformed using the predefined reversible matrix A and binary bitmask Λ to map it to a latent space for data processing. Next, Gaussian noises are added to it, and moment t is selected by a predefined program. In practice, each moment in $\{t_1, t_2 \cdots t_N\}$ is determined one by one using the ternary search method, which can obtain higher image quality. This is consistent with previous studies [24]. Additionally, the image added with Gaussian noises is sampled to obtain a sample x, and then the sample and the corresponding noise addition time are jointly input into the consistency model f_θ for one-step sample deblurring. As mentioned above, the consistency model can significantly improve image quality and reduce the image evaluation index FID through alternating multiple adding noise and denoising steps. Therefore, though repeating steps 3–5 of Algorithm 2 multiple times, the optimized deblurred image is finally obtained and output.

Experiments

This section first presents the basic experiment setup. Then, ablation experiments are conducted on each component of ID-CDM, and the iteration step size in image deblurring, distillation model performance, and two types of numerical calculators are compared and analyzed. Additionally, the ID-CDM method is evaluated on public datasets, including GoPro [7], HIDE [80], RealBlur-R, and RealBlur-J [81], and it is quantitatively and qualitatively compared with existing methods.

The GoPro [7] blurred dataset consists of 3214 blurred images with a size of 1280 × 720 pixels, including 2103 training images and 1111 testing images. The dataset comprises realistically blurred images captured by a camera and corresponding ground truth images. Compared with synthetic blurred images using blur kernels, the GoPro dataset contains more realistic and complex blurs. In this experiment, the proposed ID-CDM model tested on other datasets is trained on the GoPro training dataset.

Experimental setup

When training the ID-CDM distillation model, the U-net network architecture is adopted. As for model parameters, a batch size of 64 is used on the GoPro dataset. Additionally, in this study, the patch size and batch size for each stage are configured according to a progressive training strategy, specified as follows: [(128, 64), (128, 10), (192, 8), (256, 4), (320, 2), (384, 2)]. The Adam [82] optimizer is utilized to train the network architecture, the momentum is set to (0.9,0.9999), the weight decay is set to 1e-2, the learning rate is set to 1e-4, the value of ξ is set to 25, and the number of diffusion steps is set to 4000. A progressive training strategy was employed, with a total of 300 K iterations divided into six stages: [20 K, 20 K, 20 K, 20 K, 20 K, 200 K]. When training the ID-CDM consistency model, the network architecture from the literature [18] is used in the experiment on the GoPro dataset, and the weights are smoothed through exponential moving average (EMA), which improves the quality of sampled generated images but prolongs the training time. In this experiment, the EMA decay rate is set to 0.9999, the metric matrix $d(\cdot, \cdot)$ of the objective function is set to $d(x, y) = \|x - y\|_2^2$, the imnoising moment T of the prior distribution is set to 80, and the number of intervals Nin $[\varepsilon, T]$ is set to 60. Other training parameter settings are listed in Table 1. In this study, four NVIDIA A800 GPUs, each with 80 GB of video memory, were utilized to perform the deblurring training of ID-CM on the GoPro dataset. The software environment for the experiments was configured as follows: CUDA version 11.3, PyTorch version 1.9.0, and Python version 3.9.

This study employs the PSNR and SSIM evaluation metrics to assess the performance of ID-CDM. PSNR is a widely adopted metric for measuring image fidelity. It quantifies the difference between the ground truth and the generated image in terms of pixel-wise error. Higher PSNR values indicate better image quality; SSIM evaluates the structural similarity between two images, which is essential in image restoration tasks where fine details and texture preservation are important. SSIM offers a perceptually more relevant measure than PSNR, as it incorporates comparisons of luminance, contrast, and structural information, rather than just absolute pixel differences.

The zero-shot image deblurring task is implemented following the pseudo-code in Algorithm 2. Similarly, N is set to 60, and the orthogonal matrix P is obtained through the QR decomposition of the identity matrix. If the sum of the elements in the first column of the Q matrix is negative, the Q matrix is taken as negative and used as the orthogonal matrix P.

Ablation studies

Ablation studies are conducted to analyze the proposed model. First, the number of iteration steps N of image deblurring under zero-shot conditions is investigated. Then, the performances of using DCSM and EDM [26] as the distillation model are compared. Finally, the calculation methods using the two numerical calculators are compared, and their respective impacts on the experimental results are analyzed.

(1) When deblurring images with the trained ID-CDM under zero-shot conditions, the hyperparameter N (the number of iterative sampling steps) in Algorithm 1 should be determined. In this image deblurring experiment on the GoPro dataset, the impacts of different values of N on PSNR and the zero-shot image generation time are compared. As listed in Table 2, the smaller the value of N, the faster the image deblurring process by ID-CD, and the shorter the convergence time, but the lower the quality of the generated images. When the number of iterative sampling steps N is set to 60, the time used for image deblurring is 5.8 s, and the PSNR is 33.19 dB, which is the optimized value in the table.

(2) To verify the performance of the proposed score model, the two distillation models and numerical calculators are combined for experiments in the image generation stage, and the experiments take samples from the CIFAR-10 dataset. As listed in Table 3, the quality of the generated images by the consistency independence model (CT) [24] without a distillation model is relatively low, while ID-CDM using DCSM as the distillation model and using the Euler or Heun numerical calculator [26] both outperforms CT, and the results of using the Euler numerical calculator are slightly better than those of using the Heun numerical calculator. Additionally, paper [24] using EDM as the distillation model and combining it with the Heun numerical calculator achieves even better results, leading to FID (Fréchet Inception Distance) as low as 3.55 and also higher image generation quality.

(3) In practice, the drift term primarily shifts the data distribution along low-frequency components and can be effectively absorbed by the network parameterization. Table 4 summarizes the comparison of different model variants. Starting from EDM with drift, we first replace the drift term with our proposed DCSM (ID-DDPM). This modification reduces FLOPs from 170.2 G to 135.4 G and inference time from 9.6 to 9.1 s, demonstrateing that omitting the drift does not harm perceptual quality or PSNR/SSIM, and reduces complexity. As the network no longer needs to additionally predict drift.

Additional, to assess the impact of the consistency model, we compare the full ID-CDM model (with DCSM and consistency model) against a version that uses only DCSM without the consistency model, namely ID-DDPM (see the second and third rows of Table 4). This paper employs DDPM (denoising diffusion probabilistic models) as a replacement

Table 1 The hyperparameter settings for training ID-CDM on the GoPro dataset

Learning rate	EMA decay rate	N	ε	Training iterations	ODE solver
4e−4	0.9999	60	0.001	400K	Euler
Batch size	Number of GPUs	$d(\cdot,\cdot)$	ρ	Dropout probability	T
1024	4	l_2	7	0.0	80

Table 2 Number of zero-shot iterative sampling steps by GoPro

Steps	N = 40	N = 50	N = 60	N = 70	N = 80
PSNR/dB	31.24	31.93	**33.19**	32.98	33.05
Time/s	**3.2**	3.9	5.8	7.5	9.7

The boldface numbers in all tables highlight the optimal results

Table 3 Comparative experiments of distillation models and numerical calculators

Method	Distillation model		ODE solver		FID ↓
	DCSM	EDM	Euler	Heun	
ID CDM-Euler (ours)	✔	×	✔	×	4.02
CD-Euler [24]	×	✔	✔	×	6.54
CD-Heun [24]	×	✔	×	✔	**3.55**
ID CDM-Heun (ours)	✔	×	×	✔	4.26
CT-Heun [24]	×	×	×	✔	8.70

The boldface numbers in all tables highlight the optimal results

for the consistency model. We measure the inference time and deblurring quality to demonstrate the contribution of the consistency model to both efficiency and image quality. As indicated by the comparison between rows 2 and 3 in Table 4, replacing consistency model with DDPM leads to a significant drop in SSIM, suggesting that the consistency model is important for maintaining structural integrity. ID-CDM achieves the highest PSNR (33.19) and SSIM (0.970), while further reducing FLOPs to 127.5 G and inference time to 5.8 s per image, a 40% acceleration compared to EDM. Without the consistency model, ID-CDM fails to preserve high-frequency details as effectively, which degrades image quality.

Furthermore, we reduced the experimental scale and compared the convergence speed among three model variants: EDM + DDPM, ID-CDM (DCSM + consistency model), and ID-DDPM (DCSM + DDPM). As shown in the Fig. 5, it can be observed from the table that ID-CDM and ID-DDPM exhibit nearly identical convergence rates, which are significantly faster than that of the EDM + DDPM combination. This further validates that DCSM (without drift) achieves better convergence performance compared to DCSM (with drift). The result aligns with our hypothesis that, empirically, the absence of the drift term does not hinder convergence; instead, it accelerates training and enhances stability.

Figure 6 presents a visual comparison of the three model variants listed in Table 4. It can be observed that in regions with severe high-frequency text blur, the ID-CDM method exhibits clearer character structures compared to ID-DDPM and EDM + DDPM, and produces texture features closer to

Table 4 Ablation study on the effect of the drift term and Consistency model

Model variant	PSNR ↑	SSIM ↑	FLOPs(G)↓	Inference time (s/img)↓
EDM (with drift) + DDPM	33.08	0.964	170.2	9.6
ID-DDPM: DCSM (without drift) + DDPM	33.15	0.965	135.4	9.1
ID-CDM: DCSM (without drift) + consistency model	**33.19**	**0.970**	**127.5**	**5.8**

The boldface numbers in all tables highlight the optimal results

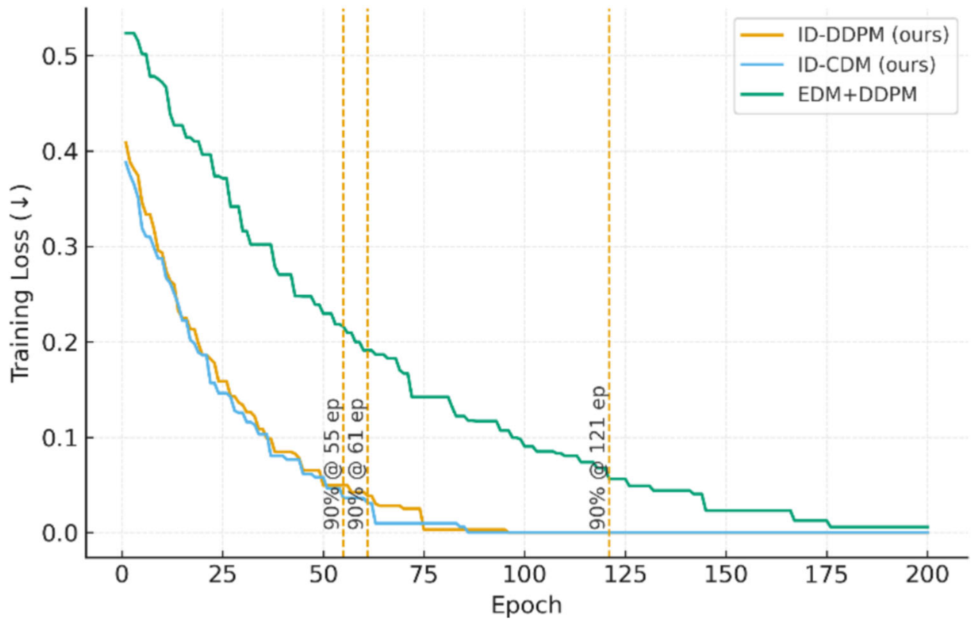

Fig. 5 Convergence trend: ID-CDM, ID-DDPM, and EDM + DDPM

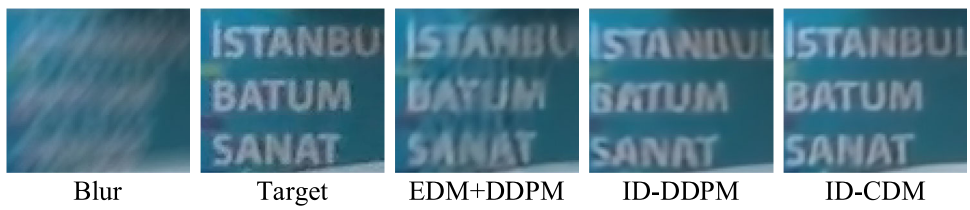

Fig. 6 The qualitative results of model variant

the ground-truth image. This further validates that the integration of DCSM with the consistency model enhances the detailed reconstruction capability of deblurred images.

Quantitative analysis

The proposed ID-CDM model is applied to zero-shot image deblurring on the GoPro dataset, and the experimental results are compared with those of existing image deblurring methods, including deblurring methods based on convolutional neural network (DeepDeblur [7], GAMD [83]), GAN-based image deblurring models (DeblurGAN [32], DeblurGAN-v2 [33], and DBGAN [30]), transformer-based deblurring network models (Uformer-B [11], Stripformer [13], and Restormer [12]), and deblurring methods derived from diffusion models (IR-SDE [84], DiffIR [62], and HI-Diff [85]).

Table 5 shows the comparison results of ID-CDM and the above-mentioned methods on four deblurring datasets. Without separately training the dataset for deblurring under zero-shot conditions, ID-CDM achieves a PSNR of 33.19 dB and an SSIM [86] of 0.970 on the GoPro dataset, which generally outperforms the models derived from CNN, GAN [19], and transformer [10]. Only on the HIDE dataset [80], the SSIM of the proposed model is slightly inferior to that of the Uformer-B model [11]. Compared to diffusion-based image deblurring methods, ID-CDM outperforms IR-SDE [84] by 2.49 dB in terms of PSNR on the GoPro dataset, and underperforms the HI-Diff method [85] by 0.14 dB in terms of PSNR but outperforms it in terms of SSIM. On the HIDE [80] and RealBlur [81] datasets, the ID-CDM method achieves similar image quality to the HI-Diff method [85], demonstrating excellent deblurring performance on both synthetic and realistic blurred datasets.

To evaluate the perceptual quality of the ID-CDM deblurring method, this study compares it with other methods using the LPIPS metric. LPIPS measures perceptual similarity between images using a deep neural network, making it more aligned with human perception. This metric is particularly valuable in evaluating perceptual quality, as it accounts for high-level features like texture and object coherence. The experimental results are shown in Table 6. It shows that ID-CDM outperforms existing methods in LPIPS, indicating that ID-CDM not only restores high-fidelity images but also produces perceptually realistic images that align more closely with human perception. This highlights the strength of ID-CDM in generating images that are both high-quality and visually natural.

To verify the zero-shot data processing ability of ID-CDM, experiments are also carried out on the CelebA-1 K

Table 5 The quantitative results on deblurring datasets for testing

Method	GoPro		RealBlur-R		RealBlur-J		HIDE	
	PSNR ↑	SSIM ↑	PSNR ↑	SSIM ↑	PSNR ↑	SSIM ↑	PSNR ↑	SSIM ↑
DeepDeblur [7]	29.08	0.914	32.51	0.841	27.87	0.827	25.73	0.874
GAMD [83]	33.14	0.9284	34.00	0.9265	–	–	–	–
DeblurGAN [32]	28.70	0.858	33.79	0.903	27.97	0.834	24.51	0.871
DeblurGAN-v2 [33]	29.55	0.934	35.26	0.944	28.70	0.866	26.61	0.875
DBGAN [30]	31.10	0.942	33.78	0.909	24.93	0.745	28.94	0.915
Uformer-B [11]	32.97	0.967	36.22	0.957	29.06	0.884	30.83	**0.952**
Stripformer [13]	33.08	0.962	36.08	0.954	28.82	0.876	31.03	0.940
Restormer [12]	32.92	0.961	36.19	0.957	28.96	0.879	31.22	0.942
IR-SDE [84]	30.70	0.901	33.96	0.918	24.21	0.729	–	–
DiffIR [62]	33.20	0.963	–	–	–	–	31.55	0.947
HI-Diff [85]	**33.33**	0.964	36.28	**0.958**	**29.15**	**0.890**	31.46	0.945
ID-CDM (ours)	33.19	**0.970**	**36.34**	0.955	28.96	0.887	**31.53**	0.950

The boldface numbers in all tables highlight the optimal results

Table 6 Performance comparison with perceptual quality metrics

Method	Restormer [12]	Uformer [11]	Stripformer [13]	HI-Diff [85]	DiffIR [62]	ID-CDM(ours)
LPIPS	0.0841	0.0868	0.0771	0.0799	0.0787	**0.0768**

The boldface numbers in all tables highlight the optimal results

Table 7 The quantitative results of the CelebA-1 K zero-shot image deblurring diffusion model

Method	CelebA-1 K			Time (s/image)	Inference steps
	PSNR ↑	SSIM ↑	FID ↓ [88]		
DDNM [47]	**27.25**	**0.782**	**62.65**	16.2	100
DDRM [46]	27.21	0.767	87.32	8.9	20
DPS [48]	25.56	0.688	65.67	130.1	1000
ID-CDM (ours)	26.15	0.745	71.68	**5.3**	**10**

The boldface numbers in all tables highlight the optimal results

dataset [87]. Under the same configurations and parameters, the proposed ID-CDM is compared with diffusion-based DDNM [47], DDRM [46], and DPS [48] models under zero-shot conditions. As presented in Table 7, although ID-CDM underperforms the existing state-of-the-art model DDNM [47] in terms of some evaluation metrics, ID-CDM achieves a PSNR of 26.15 dB in only 10 iteration steps, which is 0.59 dB higher than that of DPS [48] using 1000 iteration steps for deblurring. Among the models listed in Table 7, ID-CDM has the shortest sampling time, about 5 s on average per image. Compared to the DDRM model [46], ID-CDM not only generates images faster but also performs better in the perceptual metric (FID [88]). Therefore, in the case of image deblurring, ID-CDM not only significantly reduces the sampling steps and processing time compared to diffusion models but also exhibits excellent deblurring performance.

Visual results

The performances of the proposed ID-CDM and other methods on the GoPro testing set and the HIDE [80] testing set are compared qualitatively. As illustrated in Figs. 7 and 8, ID-CDM can effectively restore blurred images from synthetic datasets, especially in areas with high texture and fine granularity such as text and vehicles. According to Fig. 8, both

Fig. 7 The qualitative results on the GoPro testing set

Fig. 8 The qualitative results on the HIDE dataset

HI-Diff [85] and ID-CDM better display text information in blurred images, leading to better visual deblurring effects.

Figure 9 provides a qualitative comparison of the performances of ID-CDM and other methods on the RealBlur-J testing set [81]. According to Table 5, the quantitative results of various methods on this dataset are inferior to those on the GoPro and HIDE datasets. This is because the real-world dataset RealBlur-J contains various mixed blurs, as the dataset is usually collected in mixed, moving, and blurred environments, resulting in complex and nonlinear blurs. As a result, the deblurring tasks are more challenging. As demonstrated in Fig. 9, the images restored by most methods have serious gray pseudo-shadows or still contain significant blurs. Compared with other methods, ID-CDM can more accurately restore image textures, the edges of text are clearer, and the images obtained by ID-CDM are more similar to the real images (GT) under zero-shot conditions, demonstrating the excellent deblurring performance of ID-CDM.

Compared to the RealBlur-J dataset, the RealBlur-R dataset mainly comprises images taken in low-light nighttime scenes, where insufficient illumination and diverse blur effects pose greater challenges. These include a combination of factors such as low-light noise, handshake-induced motion blur, and interference from complex artificial light sources. Figure 10 presents a visual comparison between the ID-CDM method and other state-of-the-art deblurring algorithms on the RealBlur-R dataset. It can be clearly observed that in regions with high-frequency luminance variations, such as text areas (e.g., signs and labels), the blur is particularly severe due to the combined effects of uneven illumination and motion blur during nighttime capture. These compared methods still exhibit noticeable residual blur and artifacts in these regions, indicating limited recovery performance. In contrast, ID-CDM demonstrates a clearer restoration of textual edge structures and fine details, with results that are visually closer to the ground-truth images. This demonstrates that ID-CDM attains greater perceptual authenticity while maintaining fine structural details more effectively.

In addition to qualitative testing on four public blur datasets, this paper also evaluates ID-CDM against Restormer [12] and GAMD [83] on our privately constructed dataset, MSB[1] (machine shop blur), with qualitative results presented in Fig. 11. It can be observed that, compared to other methods, ID-CDM achieves superior deblurring performance in high-frequency text regions, producing results

[1] https://drive.google.com/file/d/1F8ymfSwq5TLhiMzp2FiNUuwL19Mlxnaj/view?usp=sharing.

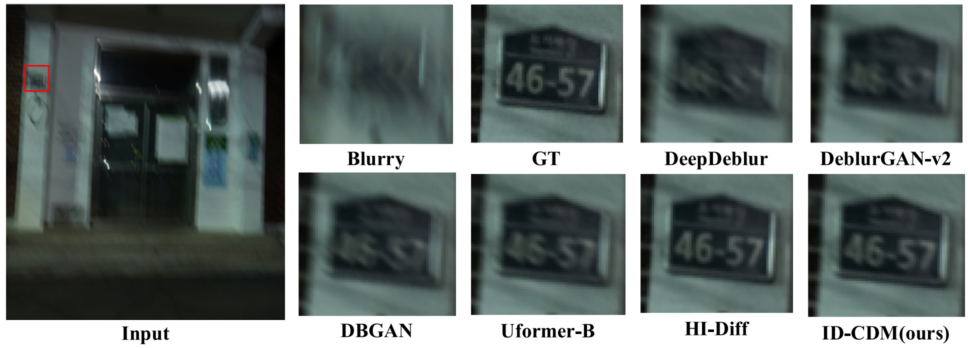

Fig. 9 The qualitative results on the RealBlur-J dataset

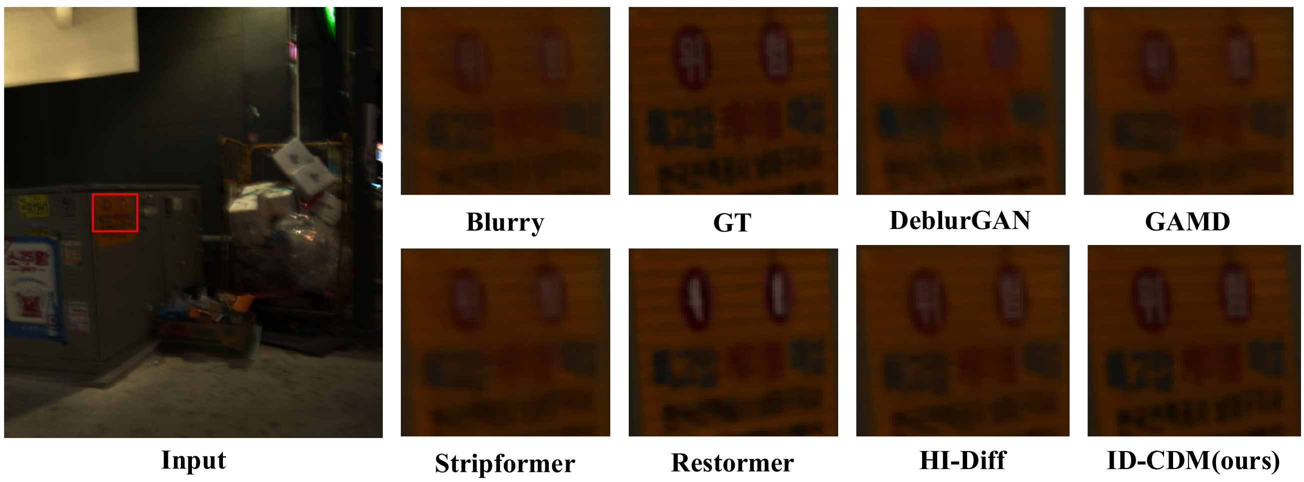

Fig. 10 The qualitative results on the RealBlur-R dataset

Fig. 11 The qualitative results on the MSB dataset

closer to the ground-truth sharp images. This finding is consistent with the results obtained on the four public blur datasets.

To further enhance the interpretability of the proposed ID-CDM model, we conduct a feature-level comparison of deblurred images produced by ID-CDM, HI-Diff [85], and Restormer [12]. Instead of relying solely on pixel-based metrics such as PSNR and SSIM, we analyze edge and frequency-domain characteristics of the restored images to evaluate structural fidelity and high-frequency recovery.

For each deblurred image, we extract and visualize three types of feature representations: (1) Sobel Edge Map—captures the gradient intensity along horizontal and vertical directions, reflecting the sharpness of edges. (2) Canny Edge Map—provides a more refined edge representation, highlighting fine structural details such as text and object contours. (3) Frequency spectrum (FFT magnitude)—reveals the distribution of spatial frequencies in the image, where stronger high-frequency components indicate clearer texture and detail recovery.

As shown in Fig. 12, ID-CDM produces sharper Sobel and Canny edge responses, particularly in fine text regions and object boundaries. Moreover, the FFT spectrum of the deblurred images generated by ID-CDM contains more pronounced high-frequency energy, further indicating effective restoration of fine details. In contrast, both HIdiff [85] and Restormer [12] yield relatively smoother results with suppressed noise, but at the cost of losing high-frequency details in the deblurred images. The Sobel and Canny maps of these two methods also exhibit blurred edges, while their FFT spectra demonstrate reduced high-frequency responses, suggesting weaker texture preservation. The comparison confirms that ID-CDM not only achieves higher pixel-level quality but also restores clearer edge structures and preserves richer high-frequency information. This feature-level analysis provides additional evidence of ID-CDM's superiority in

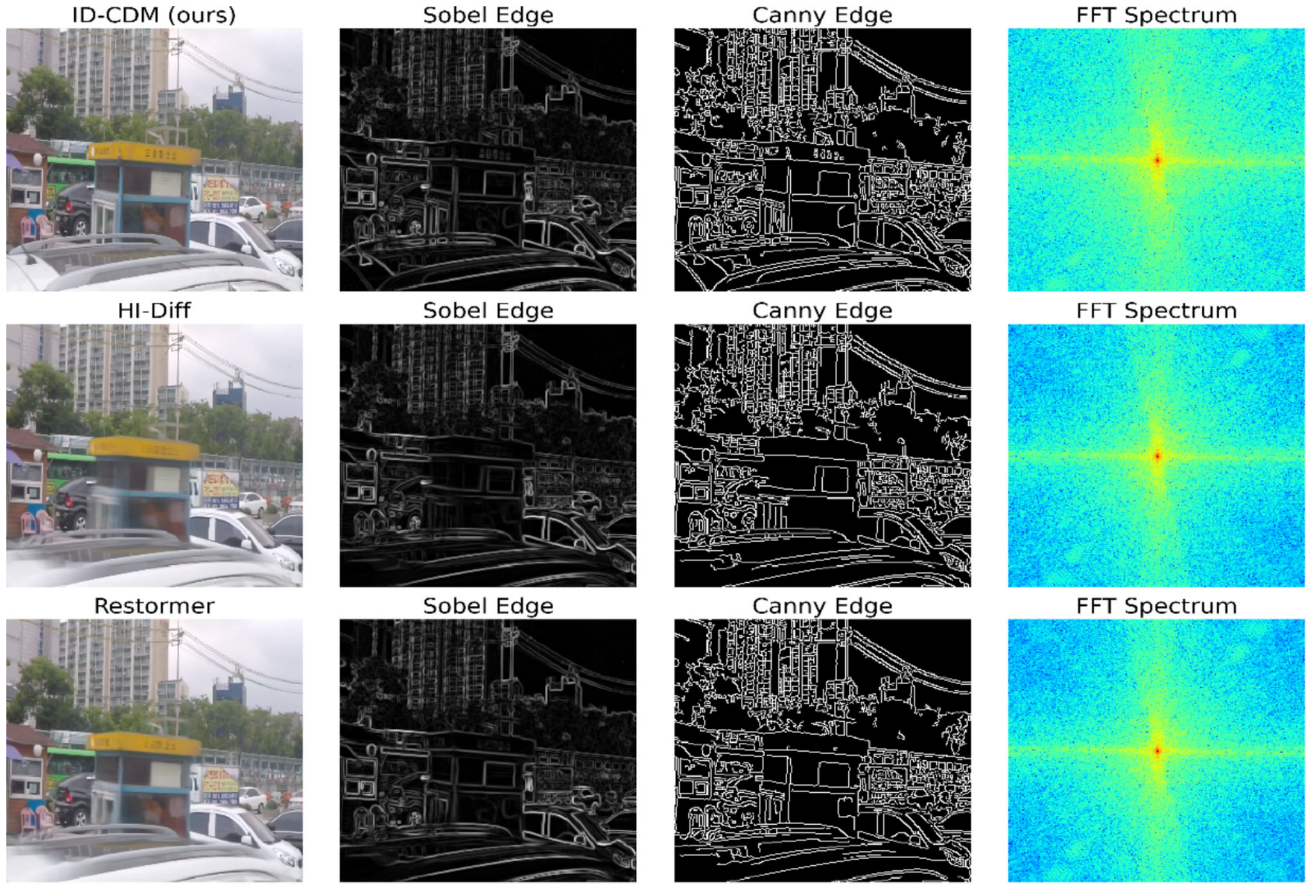

Fig. 12 Feature map comparison of deblurred results from different methods

handling fine details and complex textures, which are crucial for real-world deblurring applications.

Overall, the qualitative results demonstrate that ID-CDM can effectively deblur images in both synthetic and realistic scenes, and compared to other algorithms, it can generate images that are clearer and visually closer to real images.

Computational efficiency analysis

In addition to restoration quality, computational efficiency is a crucial factor for the practical deployment of deblurring models. To provide a comprehensive analysis, we compare ID-CDM with CNN-based, transformer-based, and diffusion-based deblurring methods in terms of FLOPs, inference speed, and PSNR. For fairness, all experiments are conducted using four NVIDIA A800 GPUs with input images resolution of $3 \times 256 \times 256$.

As shown in Table 8, CNN-based and transformer-based approaches such as Uformer-B [11], Stripformer [13], and Restormer [12] exhibit relatively low training cost and fast inference, but they often sacrifice detail recovery in complex blur scenarios. In contrast, diffusion-based methods such as IR-SDE and HI-Diff achieve superior deblurring quality but require extremely high computational cost and low inference speed due to hundreds of iterative sampling steps.

ID-CDM strikes a balance between efficiency and accuracy. Benefiting from the diffusion coefficient-dominated score model (DCSM) and its integration into the consistency model, ID-CDM reduces FLOPs to 127.5 G, achieves an inference time of about 5.8 s per image (256×256), which is significantly faster and lighter than IR-SDE ($\sim$ 25 s) and HI-Diff (~ 15 s). While the training time of ID-CDM is slightly longer than lightweight transformer-based models due to the distillation stage, the one-time training expense is compensated by its efficient zero-shot inference capability.

These results demonstrate that ID-CDM achieves a favorable trade-off between computational efficiency and deblurring performance.

Robustness analysis

To further validate the practical applicability of ID-CDM, we analyze its robustness against different types of noise and combined image degradations. In real-world scenarios, blurred images are often contaminated by additional distortions such as sensor noise, low illumination, or compression

Table 8 Computational efficiency comparison

Method	FLOPs(G)	Inference time (s/img)	PSNR	SSIM
Uformer-B [11]	155.8	0.12	32.97	0.967
Stripformer [13]	140.5	**0.10**	33.08	0.962
Restormer [12]	165.2	0.15	32.92	0.961
IR-SDE [84]	210.3	25.0	30.70	0.901
HI-Diff [85]	142.6	15.2	**33.33**	0.964
ID-CDM (ours)	**127.5**	5.8	33.19	**0.970**

The boldface numbers in all tables highlight the optimal results

artifacts. Therefore, we evaluate ID-CDM under one representative robustness settings: additive Gaussian noise with different standard deviations.

We follow the GoPro datasets for evaluation. Gaussian noise with standard deviations $\sigma = 5, 15, 25$ is added to the blurred input images. All methods are retrained using their official implementations and then tested under the same degraded conditions. Performance is measured using PSNR and SSIM. As shown in Table 9, transformer-based models (Uformer [11], Restormer [12]) show significant performance degradation as the noise level increases, while diffusion-based methods (HI-Diff [85], ID-CDM) maintain better stability. Notably, ID-CDM demonstrates the smallest drop in PSNR/SSIM when noise level increases, confirming that the integration of DCSM with consistency learning enhances robustness against noise.

Discussion

Existing deblurring approaches exhibit different limitations: CNN-based and GAN-based methods are fast but struggle with recovering fine details in complex real-world blurs; transformer-based architectures capture long-range dependencies but often incur high parameter complexity; diffusion-based methods deliver superior perceptual quality yet suffer from extremely high sampling costs and slow inference; and consistency models, though efficient, generally fail to restore high-frequency details when directly applied to image restoration tasks. In contrast, our proposed ID-CDM addresses these limitations from three perspectives: (1) by adopting a diffusion coefficient-dominated score model (DCSM), the computational complexity and instability of diffusion models are significantly reduced; (2) by integrating DCSM into the consistency framework, ID-CDM preserves high-frequency image details while retaining few-step sampling efficiency; and (3) by leveraging latent-space zero-shot processing, ID-CDM eliminates the need for paired training data and demonstrates strong generalization across synthetic and real-world datasets. These advantages highlight the novelty of ID-CDM and provide new insights into designing efficient and effective diffusion-based restoration frameworks.

Beyond synthetic and benchmark datasets, the practical deployment of image deblurring models requires careful consideration of real-world scenarios. The proposed ID-CDM model, with its efficiency and robustness, has promising potential for a variety of applications: (1) Modern smartphone cameras often suffer from motion blur due to hand shake or low-light shooting. ID-CDM could be integrated into camera pipelines to enhance photo quality in real time. (2) Blurred images from on-board cameras (e.g., due to fast vehicle motion or adverse weather) can reduce the accuracy of perception systems. ID-CDM can improve the clarity of visual inputs, thereby enhancing safety in autonomous navigation. (3) Medical imaging: certain imaging modalities may suffer from motion blur (e.g., patient movement during scans). ID-CDM can help improve the clarity of such images, supporting more accurate diagnostics. (4) Video surveillance and security: surveillance cameras frequently capture blurred frames due to subject motion or low-light environments. Applying ID-CDM could significantly improve the readability of these frames, aiding in recognition and analysis.

Despite these strengths, ID-CDM also has several limitations. (1) Despite achieving efficient inference, the training process of ID-CDM remains computationally expensive due to the distillation step and the integration of DCSM within the consistency framework. Future research could explore more lightweight distillation strategies, possibly incorporating self-supervised learning techniques, to reduce the overall training cost. Additionally, using more efficient optimizers or training procedures could significantly reduce the training time. (2) While ID-CDM performs well across various types of blur and noise, its performance can degrade when dealing with extreme degradations, such as severe motion blur combined with high noise levels or under low-light conditions. This is a common challenge in real-world applications, and future improvements could focus on enhancing the model's robustness under such severe degradation scenarios. Methods like multi-scale deblurring or adaptive refinement could help address these challenges. (3) ID-CDM reduces computational complexity compared to traditional diffusion models,

Table 9 Robustness evaluation under Gaussian noise (GoPro dataset, σ = 0, 5, 15, 25)

Method	PSNR/SSIM			
	σ = 0 (clean blur)	σ = 5	σ = 15	σ = 25
Uformer-B [11]	32.97/0.967	30.21/0.941	26.85/0.892	23.41/0.812
Restormer [12]	33.45/0.969	30.62/0.944	27.31/0.901	23.87/0.828
HI-Diff [85]	**33.33**/0.964	31.95/0.952	29.20/0.922	26.45/0.880
ID-CDM (ours)	33.19/**0.970**	**32.02/0.957**	**29.56/0.930**	**26.83/0.889**

The boldface numbers in all tables highlight the optimal results

but its memory usage can still be relatively high, especially during inference. For real-time applications or deployment on edge devices, further optimization of the model's memory footprint and computational efficiency is essential. Techniques such as model pruning, quantization, or leveraging lightweight architectures could be explored to reduce memory usage without sacrificing performance.

Building upon the current work, future research could explore the following areas: (1) Incorporating multi-scale information could improve the model's ability to restore details at different levels of resolution, especially for complex real-world scenes with varying levels of degradation. (2) While ID-CDM has been validated on low-light blur scenarios via the RealBlur-R dataset, its generalization to spatially varying distortions such as rolling shutter artifacts remains unexplored and will be pursued in future work. (3) Future work could investigate adaptive sampling strategies to dynamically adjust the number of diffusion steps based on the severity of degradation, reducing computational overhead while maintaining image quality. (4) In addition, the current framework has not been fully optimized for ultra-high-resolution inputs (e.g., 4 K or larger), where memory overhead becomes significant. Another open challenge lies in extending the method to real-time video deblurring, which requires maintaining both temporal consistency and low latency. Potential solutions include adaptive iteration schedules, patch-based memory-efficient processing, and model distillation toward lightweight architectures. This paper considers these promising directions for future research.

Conclusion

This paper proposes the ID-CDM model for image deblurring. It integrates the diffusion coefficient-dominated score model (DCSM) as a distillation model. The iterative sampling of ID-CDM not only improves the quality of image generation, but more importantly, it greatly overcomes the defects of the diffusion model in terms of many iteration steps, long time, and large computational cost. Meanwhile, this paper performs image deblurring on both synthetic blur datasets and real scene blur datasets under zero-shot conditions. Invertible linear transformation is employed to compress the blurred images into latent space, and ID-CDM is used as a denoising network to implement the deblurring process. The proposed ID-CDM reduces the computational complexity of the model while reducing additional artifacts of deblurred images that do not exist in the given clear image, achieving generalization in complex blurred scenes. The extended experiments on synthetic and real-world blur datasets indicate that our proposed ID-CDM outperforms current methods.

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Declarations

Conflict of interest The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Ethical and informed consent for data used All data used in this paper conforms to the Ethical and informed consent specification.

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